

Week4

DESCRIBE Books;

The screenshot shows a database management tool interface. At the top, there's a toolbar with various icons and a 'Limit to 1000 rows' dropdown. Below the toolbar, a SQL editor contains the following code:

```
1  USE bookflow_db;
2  CREATE TABLE Books
3  (
4      book_id INT AUTO_INCREMENT PRIMARY KEY,
5      title VARCHAR(200) NOT NULL,
6      author VARCHAR(150) NOT NULL,
7      genre VARCHAR(100),
```

Below the editor, there's a 'Result Grid' section. It includes a 'Filter Rows' input, an 'Export' button, and a 'Wrap Cell Content' checkbox. The grid displays the following table structure:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	
embedding	json	YES		NULL	

Below the grid, there's a 'Result 1' tab and a 'Read Only' status. The 'Output' section shows a list of actions:

#	Time	Action	Message
1	15:00:27	USE bookflow_db	0 row(s) affected
2	15:00:27	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCH...	Error Code: 1050. Table
3	15:00:28	DESCRIBE Books	7 row(s) returned

Insert

The screenshot shows a database management tool interface. At the top, there's a toolbar with various icons and a 'Limit to 1000 rows' dropdown. Below the toolbar, a SQL editor contains the following code:

```
4  INSERT INTO Books ( title, author, genre, description, published_year ) VALUES ...
```

Below the editor, there's a 'Result Grid' section. It includes a 'Filter Rows' input, an 'Export' button, and a 'Wrap Cell Content' checkbox. The grid displays the following table structure:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	
embedding	json	YES		NULL	

Below the grid, there's a 'Result 1' tab and a 'Read Only' status. The 'Output' section shows a list of actions:

#	Time	Action	Message
4	15:02:43	INSERT INTO Books (title, author, genre, description, published_year) VALUES ...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

SELECT

```
book_id,
title,
author,
genre,
published_year
```

FROM Books

ORDER BY book_id;

SQL Editor interface showing a query and its results.

```

92  book_id,
93  title,
94  author,
95  genre,
96  published_year
97  FROM Books
98  ORDER BY book_id;

```

Result Grid:

book_id	title	author	genre	published_year
1	The Martian	Andy Weir	Science Fiction	2011
2	Dune	Frank Herbert	Science Fiction	1965
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4	Pride and Prejudice	Jane Austen	Romance	1813
5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
6	Foundation	Isaac Asimov	Science Fiction	1951
7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
8	Enders Game	Orson Scott Card	Science Fiction	1985

DESCRIBE Books;

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	
embedding	json	YES		NULL	

UPDATE Books

SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)

WHERE book_id = 1;

9 15:08:40 UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1 0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0

UPDATE Books

SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)

WHERE book_id = 2;

10 15:09:24 UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2 0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0

UPDATE Books

SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)

WHERE book_id = 3;

✓ 11 15:10:24 UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3 0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0

UPDATE Books

SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)

WHERE book_id = 4;

✓ 12 15:10:58 UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4 0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0

UPDATE Books

SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)

WHERE book_id = 5;

✓ 13 15:11:31 UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5 0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0

UPDATE Books

SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)

WHERE book_id = 6;

✓ 14 15:12:00 UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6 0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0

UPDATE Books

SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)

WHERE book_id = 7;

✓ 15 15:12:23 UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7 0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0

UPDATE Books

SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)

WHERE book_id = 8;

✓ 16 15:12:57 UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8 0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0

UPDATE Books

SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)

WHERE book_id = 9;

✓ 17 15:13:26 UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9 0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0

UPDATE Books

SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)

WHERE book_id = 10;

✓ 18 15:14:07 UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10 0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0

SELECT

book_id,

```

title,
embedding
FROM Books
ORDER BY book_id;

```

Result Grid			
		Filter Rows:	
		Edit:	
		Export/Import:	
		Wrap Cell Content	
	book_id	title	embedding
▶	1	The Martian	[0.90, 0.80, 0.20]
	2	Dune	[0.85, 0.75, 0.25]
	3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
	4	Pride and Prejudice	[0.10, 0.20, 0.90]
	5	The Great Gatsby	[0.15, 0.30, 0.80]
	6	Foundation	[0.95, 0.65, 0.20]
	7	The Hobbit	[0.30, 0.75, 0.90]
	8	Enders Game	[0.88, 0.85, 0.15]

```

SELECT
book_id,
title,
author,
genre,
ROUND(
  SQRT(
    POW(
      CAST(
        JSON_EXTRACT(embedding, '$[0]')
        AS DECIMAL(10,6)
      ) - 0.90,
      2
    )
    +
    POW(
      CAST(
        JSON_EXTRACT(embedding, '$[1]')

```

```
        AS DECIMAL(10,6)
    ) - 0.80,
    2
)
+
POW(
    CAST(
        JSON_EXTRACT(embedding, '$[2]')
        AS DECIMAL(10,6)
    ) - 0.20,
    2
),
4
) AS l2_distance
FROM Books
WHERE embedding IS NOT NULL
ORDER BY l2_distance ASC
LIMIT 3;
```

Automatic context manually get help to

```

173      4
174
175    ) AS l2_distance
176 FROM Books
177 WHERE embedding IS NOT NULL
178 ORDER BY l2_distance ASC
179 LIMIT 3;

```

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0735
2	Dune	Frank Herbert	Science Fiction	0.0866

Result 6 x Read Only Context Help Snippets

Output

#	Time	Action	Message
✓ 15	15:12:23	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	0 row(s) affected Rows matched: 1 Changed: 0 Warnings:
✓ 16	15:12:57	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	0 row(s) affected Rows matched: 1 Changed: 0 Warnings:
✓ 17	15:13:26	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	0 row(s) affected Rows matched: 1 Changed: 0 Warnings:
✓ 18	15:14:07	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	0 row(s) affected Rows matched: 1 Changed: 0 Warnings:
✓ 19	15:14:39	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	20 row(s) returned
✓ 20	15:18:26	SELECT book_id, title, author, genre, ROUND(SQRT(POW(...	3 row(s) returned

SELECT

book_id,

title,

author,

genre,

ROUND(

(

CAST(

JSON_EXTRACT(embedding, '\$[0]')

AS DECIMAL(10,6)

) * 0.90

+

CAST(

JSON_EXTRACT(embedding, '\$[1]')

AS DECIMAL(10,6)

) * 0.80

+

```

CAST(
  JSON_EXTRACT(embedding, '$[2]')
  AS DECIMAL(10,6)
) * 0.20
)
/
(
  SQRT(
    POW(
      CAST(
        JSON_EXTRACT(embedding, '$[0]')
        AS DECIMAL(10,6)
      ),
      2
    )
    +
    POW(
      CAST(
        JSON_EXTRACT(embedding, '$[1]')
        AS DECIMAL(10,6)
      ),
      2
    )
    +
    POW(
      CAST(
        JSON_EXTRACT(embedding, '$[2]')
        AS DECIMAL(10,6)
      ),
      2
    )
  )

```

```

    )
    *
    SQRT(
        POW(0.90, 2)
        +
        POW(0.80, 2)
        +
        POW(0.20, 2)
    )
),
4
) AS cosine_similarity
FROM Books
WHERE embedding IS NOT NULL
ORDER BY cosine_similarity DESC
LIMIT 3;

```

Result Grid					
Filter Rows:					
Export:					
Wrap Cell Content:					
Fetch rows:					
	book_id	title	author	genre	cosine_similarity
▶	1	The Martian	Andy Weir	Science Fiction	1
	2	Dune	Frank Herbert	Science Fiction	0.9986
	8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Result 7 x